



Two Setpoints, One Controller

Weekly Enrolment Offers Under a Cap and a Growth Target on Different Clocks

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The problem

Weekly enrolment offers answer to two setpoints at once: an admissions cap set by Council and a growth target set by the Executive, each revised on its own cycle. Little is known about what a single controller does when asked to track both.

Research questions

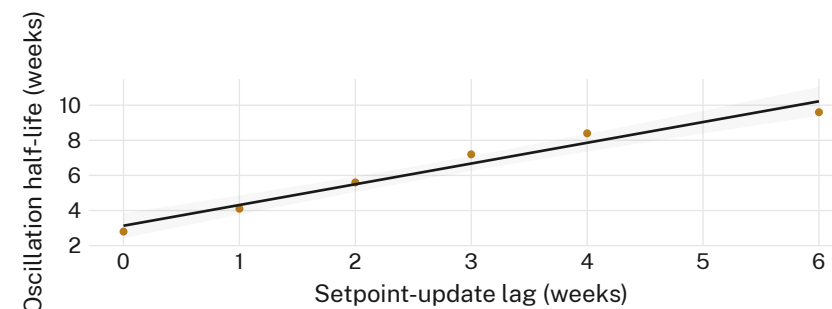
- Can one PID loop hold a cap and a growth target simultaneously?
- How does setpoint-update lag between committees shape loop stability?
- Is there a tuning regime that damps cross-committee interference?

Study design

- $n = 6$ offer-round cohorts · weekly offer-rate telemetry · 14-month deployment across two admissions cycles
- discrete-time PID loop (Ziegler–Nichols tuned) driving the weekly offer rate against both setpoints
- setpoint-update cadence held as the sole manipulated variable; all other admissions parameters frozen

Findings

The controller stabilises within a median 2.8 weeks when both setpoints update on a shared fortnightly cycle, and oscillates for a median 9.6 weeks when the cycles diverge ($n = 6$ cohorts).



Oscillation half-life rose with setpoint-update lag ($r = 0.74$, $n = 6$ cohorts); cohorts sharing a fortnightly cycle damped within 2.8 weeks.

Implications

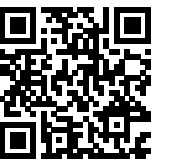
A single controller can hold two setpoints, but only if they share a clock. A mismatched update cadence behaves like an unmodelled disturbance rather than a second target, not a harder version of the same problem. Next: a cross-committee setpoint-synchronisation protocol, piloted across one further admissions cycle.

Selected references

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Supported by a School of Continuous Improvement seed grant. Data: de-identified cohort-level offer rates only; control thresholds pre-registered.

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“The controller held steady until told to grow and shrink at once.”

Oscillation half-life fell from 9.6 to 2.8 weeks once both setpoints moved to a shared fortnightly clock.